Part IV — Knowledge Evolution & Editorial Lifecycles

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1. Purpose

Clinical knowledge is not static.

Medical evidence evolves.

Educational priorities change.

Diagnostic reasoning improves.

Clinical guidelines are revised.

New misconceptions emerge.

The purpose of WEOS is therefore not simply to publish educational artifacts, but to continuously govern their evolution throughout their lifetime.

Every editorial artifact managed by WEOS progresses through a defined lifecycle.

These lifecycles establish how knowledge is created, improved, reviewed, trusted, maintained, and, when necessary, retired.

A lifecycle is not merely a workflow.

It represents the changing degree of editorial confidence placed in an artifact over time.

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2. Knowledge Evolution Philosophy

Knowledge should never be considered complete.

Publication is not the conclusion of editorial work.

Rather, publication represents the point at which institutional confidence is sufficient for learner exposure.

After publication, knowledge remains subject to:

- evidence updates
- educational improvement
- curriculum changes
- reviewer feedback
- learner performance
- AI-assisted analysis
- clinical practice changes

WEOS therefore governs living knowledge rather than static publications.

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3. Universal Editorial Lifecycle

Every governed editorial artifact follows the same conceptual progression.

```
Concept

↓

Draft

↓

Editorial Review

↓

Governance

↓

Approved

↓

Published

↓

Continuous Improvement

↓
```

```
Retired or Superseded
```

Not every artifact requires every stage.

However, every artifact progresses by increasing editorial confidence rather than increasing completeness.

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4. Editorial Confidence

The purpose of a lifecycle is to represent confidence.

Each transition increases confidence through explicit editorial decisions.

Confidence is established by evaluating:

- clinical correctness
- educational quality
- evidence sufficiency
- internal consistency
- curriculum suitability
- governance requirements

Confidence is never inferred from artifact age or publication history.

It is earned through review.

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5. Diagnosis Lifecycle

The diagnosis is the highest-level governed entity within WEOS.

Its lifecycle coordinates every subordinate editorial artifact.

```
Registry Entry

↓

Metadata Complete

↓

Educational Scaffold Required
```

```
↓

Educational Scaffold Approved

↓

Case Authoring

↓

Editorial Review

↓

Publication Review

↓

Published

↓

Maintained

↓

Retired
```

A diagnosis may remain in any stage indefinitely until sufficient editorial confidence is achieved.

Publication never bypasses governance.

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6. Educational Scaffold Lifecycle

Before comprehensive educational content can exist, every diagnosis requires an educational scaffold.

The scaffold defines the educational intent of the diagnosis.

The scaffold typically includes:

- learning objectives
- teaching rules
- differential framework
- diagnostic discriminators

- educational priorities
- clinical pearls
- pitfalls

Lifecycle:

```
Not Started

↓

AI or Human Draft

↓

Editorial Review

↓

Approved

↓

Available for Case Development

↓

Maintained
```

The scaffold precedes large-scale case generation because it establishes what future cases are intended to teach.

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7. Clinical Case Lifecycle

Clinical cases represent educational implementations of the scaffold.

Their lifecycle reflects progressive refinement of diagnostic reasoning.

```
Draft

↓

Clue Review
```

```
↓

Reasoning Review

↓

Educational Review

↓

Clinical Governance

↓

Approved

↓

Playable

↓

Published

↓

Continuous Improvement
```

A case may remain approved without being published if broader diagnosis-level publication requirements have not yet been satisfied.

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8. Knowledge Relationship Lifecycle

Knowledge relationships evolve independently of cases.

Relationships include:

- differential diagnoses
- discriminators
- supporting relationships
- prerequisite concepts
- related diagnoses

Lifecycle:

```
Suggested

↓

Evidence Assessment

↓

Editorial Review

↓

Approved

↓

Knowledge Graph

↓

Continuous Validation
```

Relationships may be strengthened, weakened, or removed as evidence evolves.

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9. Educational Content Lifecycle

Educational content supports the diagnosis outside gameplay.

Lifecycle:

```
Draft

↓

Review

↓

Approved

↓

Published
```

```
↓

Evidence Refresh

↓

Revision

↓

Republished
```

Educational revisions should preserve historical accountability while continuously improving learner understanding.

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10. AI Draft Lifecycle

Artificial intelligence generates candidate knowledge rather than governed knowledge.

AI-generated artifacts never bypass editorial review.

Lifecycle:

```
Generation

↓

Pending Human Review

↓

Approved

or

Rejected

or

Returned for Revision

↓
```

```
Governed Artifact
```

Approval transforms an AI draft into an editorial artifact.

The draft itself remains part of the editorial history.

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11. Editorial Review Lifecycle

Review itself is a governed process.

Lifecycle:

```
Review Requested

↓

In Progress

↓

Evidence Requested

↓

Decision Recorded

↓

Closed

↓

Reopened (if necessary)
```

Editorial reviews are historical records rather than transient activities.

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12. Publication Lifecycle

Publication represents institutional trust.

Lifecycle:

```
Publication Assessment

↓

Publication Approved

↓

Scheduled

↓

Released

↓

Maintained

↓

Withdrawn

or

Superseded
```

Publication status should never replace editorial status.

An artifact may remain editorially approved while awaiting publication.

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13. Continuous Maintenance

Maintenance is an explicit lifecycle stage.

Published knowledge should be continuously monitored for:

- outdated evidence
- inconsistent terminology
- educational weaknesses
- learner misconceptions
- duplicated teaching
- missing differentials

- evolving clinical guidance

Maintenance transforms publication into an ongoing editorial responsibility.

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14. Retirement

Not all knowledge remains educationally valuable forever.

Artifacts may eventually be retired due to:

- replacement
- obsolescence
- curriculum redesign
- merged diagnoses
- deprecated clinical practice
- editorial restructuring

Retirement preserves historical accountability while preventing obsolete knowledge from influencing future learners.

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15. Lifecycle Independence

Every editorial artifact possesses its own lifecycle.

A diagnosis may be published while new cases remain under review.

A teaching objective may be revised without affecting approved graph relationships.

A clinical pearl may be updated independently of case revisions.

This independence enables continuous improvement without requiring wholesale republication.

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16. Lifecycle Dependencies

Although lifecycles are independent, they remain logically connected.

Examples include:

- Educational scaffolds support case development.
- Cases reinforce learning objectives.
- Knowledge relationships strengthen differential reasoning.

- Educational content explains published knowledge.
- Evidence increases editorial confidence.
- Publication depends upon sufficient governance across all required artifacts.

Dependencies coordinate editorial progression without preventing iterative improvement.

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17. Lifecycle Principles

The lifecycle architecture of WEOS is governed by the following principles.

Every artifact has a lifecycle.

Every transition represents increased editorial confidence.

Publication is a governance milestone rather than a terminal state.

Maintenance is a permanent editorial responsibility.

Knowledge remains continuously evolvable.

Editorial history is preserved.

Independent artifacts evolve independently.

Dependencies guide progression without restricting improvement.

Closing Principle

The purpose of editorial lifecycles is not to move content through software states.

Their purpose is to govern the continuous evolution of trustworthy clinical knowledge.

Every transition within WEOS should represent a deliberate increase in institutional confidence, ensuring that medical knowledge becomes progressively more reliable, more educationally valuable, and more clinically defensible over time.